

Introduction to AI Safety, Ethics, and Society — Ch2 AI Fundamentals Quiz ANS

1. Which of the following best matches the course's working definition of “artificial intelligence”?

- A) A single universally accepted technical standard
- B) Computer systems that perform tasks typically associated with intelligence, such as problem solving, decision making, and forecasting
- C) Any computer program that processes data
- D) A branch of robotics exclusively

ANS: B) Computer systems that perform tasks typically associated with intelligence, such as problem solving, decision making, and forecasting

2. Which system is identified in the slides' AI history timeline as the “first chatbot”?

- A) Deep Blue
- B) AlphaGo
- C) ELIZA
- D) AlexNet

ANS: C) ELIZA

3. Which development is credited in the timeline as the “first neural network”?

- A) The Perceptron
- B) The Transformer
- C) ImageNet
- D) ChatGPT

ANS: A) The Perceptron

4. Which type of AI is defined in the slides as “capable of performing specific tasks only”?

- A) Narrow AI
- B) Artificial General Intelligence (AGI)
- C) Human-level AI (HLAI)
- D) Superintelligence (ASI)

ANS: A) Narrow AI

5. Which type of AI is defined as “capable of performing many cognitive tasks across multiple domains, at human or superhuman levels”?

- A) Narrow AI
- B) Artificial General Intelligence (AGI)
- C) Transformative AI (TAI)
- D) Superintelligence (ASI)

ANS: B) Artificial General Intelligence (AGI)

6. Which type of AI is defined as “capable of outclassing human performance on virtually all intellectual tasks”?

- A) Narrow AI
- B) Artificial General Intelligence (AGI)

- C) Human-level AI (HLAI)
- D) Superintelligence (ASI)

ANS: D) Superintelligence (ASI)

7. According to the slides, what is “machine learning”?

- A) Writing explicit rule-based programs for every task
- B) Developing computer systems that can learn directly from data without following explicit pre-set instructions
- C) A synonym for artificial general intelligence
- D) A hardware specification for GPUs

ANS: B) Developing computer systems that can learn directly from data without following explicit pre-set instructions

8. How do the slides define an “algorithm”?

- A) A dataset used to train models
- B) A procedure (recipe) for accomplishing a task
- C) A type of neural network layer
- D) A hardware accelerator

ANS: B) A procedure (recipe) for accomplishing a task

9. What is “training data,” as defined in the slides?

- A) The output produced by a deployed model
- B) The data from which ML models learn
- C) A model's final accuracy score
- D) A type of activation function

ANS: B) The data from which ML models learn

10. Which model access category is described as “accessible through an interface with usage restrictions, allowing users to query the model without viewing or modifying the underlying model”?

- A) Closed models
- B) API models
- C) Open models
- D) Federated models

ANS: B) API models

11. Which model access category is “fully accessible and modifiable, often part of open-source projects”?

- A) Closed models
- B) API models
- C) Open models
- D) Sandboxed models

ANS: C) Open models

12. Which computational resource is characterized in the slides by “parallel processing” and is a primary example of compute used in ML?

- A) CPU

- B) GPU
- C) Hard disk drive
- D) Network router

ANS: B) GPU

13. Which ML task is defined as “predicting categories or ‘classes’ using features of an input data point”?

- A) Regression
- B) Classification
- C) Sequence modeling
- D) Anomaly detection

ANS: B) Classification

14. Which ML task is illustrated in the slides using the example of “black swan detection”?

- A) Classification
- B) Regression
- C) Anomaly detection
- D) Generative modeling

ANS: C) Anomaly detection

15. Which evaluation metric is described in the slides as “the purity of the flagged examples”?

- A) Accuracy
- B) Precision
- C) Recall
- D) Latency

ANS: B) Precision

16. Which evaluation metric is described as “the proportion of relevant examples that were flagged”?

- A) Accuracy
- B) Precision
- C) Recall
- D) Transparency

ANS: C) Recall

17. Which type of machine learning is defined as “learning from labeled data”?

- A) Unsupervised learning
- B) Supervised learning
- C) Reinforcement learning
- D) Self-play learning

ANS: B) Supervised learning

18. Which type of machine learning is also referred to as “self-supervised learning,” since the model may generate its own supervisory signals from patterns in the data?

- A) Supervised learning
- B) Unsupervised learning

- C) Reinforcement learning
- D) Active learning

ANS: B) Unsupervised learning

19. Which type of machine learning involves an agent making decisions and improving its performance based on responses from its environment?

- A) Supervised learning
- B) Unsupervised learning
- C) Reinforcement learning
- D) Transfer learning

ANS: C) Reinforcement learning

20. According to the slides, what do deep learning (DL) models use to extract useful patterns from large quantities of data?

- A) Decision trees
- B) Neural networks: layers of interconnected nodes inspired by biological neurons
- C) Rule-based expert systems
- D) Manually engineered lookup tables

ANS: B) Neural networks: layers of interconnected nodes inspired by biological neurons

21. What is a “multi-layer perceptron (MLP),” according to the slides?

- A) A type of activation function
- B) A foundational neural network architecture with input, hidden, and output layers
- C) A method for evaluating model accuracy
- D) A dataset labeling technique

ANS: B) A foundational neural network architecture with input, hidden, and output layers

22. What is the main purpose of “residual connections” in deep learning, per the slides?

- A) To increase the model's parameter count
- B) To provide pathways for information to bypass network layers, helping preserve original content and learn hierarchical features
- C) To reduce the size of training datasets
- D) To replace activation functions entirely

ANS: B) To provide pathways for information to bypass network layers, helping preserve original content and learn hierarchical features

23. How do the slides describe “self-attention”?

- A) A method for compressing model weights
- B) A mechanism that encodes relationships between elements in a sequence by having each element “attend” to every other element
- C) A type of loss function used only in classification
- D) A hardware optimization for GPUs

ANS: B) A mechanism that encodes relationships between elements in a sequence by having each element “attend” to every other element

24. According to the slides, what is a “Transformer”?

- A) A type of activation function
- B) A deep learning architecture used in LLMs that combines self-attention and MLPs with techniques such as residual connections
- C) A dataset labeling tool
- D) A type of reinforcement learning algorithm

ANS: B) A deep learning architecture used in LLMs that combines self-attention and MLPs with techniques such as residual connections

25. What is the main takeaway from “The Bitter Lesson,” as described in the slides?

- A) Expert knowledge always outperforms scale
- B) Increasing the scale of computation and data tends to beat complex, hand-engineered research approaches in the long run
- C) Deep learning models cannot scale well
- D) Scaling laws only apply to image classifiers

ANS: B) Increasing the scale of computation and data tends to beat complex, hand-engineered research approaches in the long run